NLP.

Experiment - 05.

- **Aim:** Morphology is the study of the way words are built up from smaller meaning bearing units. Study and understand the concepts of morphology by the use of add delete table.

- **Theory:**

(1) **Introduction** -

- Morphology is the study of the internal structure of words and how words are formed using smaller meaning bearing units called morphemes.
- These morphemes include roots, prefixes, suffixes, and inflectional markers.
- Morphological analysis helps understand relationships between different forms of a word such as teach - teaching - teacher.

- In Natural language Processing (NLP), morphology is important for reducing vocabulary variation, improving text normalization and supporting tasks such as search, translation and text classification.

- One simple rule based method to represent morphological changes is the add delete table; which describes how words are transformed by removing certain characters and adding new ones.

(2). Morphemes and word formation -

* A word may consist of -

- Root morpheme - main meaning (play, act, read)
- Derivational morpheme - creates new words (teach - teacher)
- inflectional morpheme - changes grammatical form (walk - walked).

* Morphological processes include -
  - Inflection.
  - Derivation.
  - Compounding.
  - Prefixation and suffixation.

- The Add delete method mainly focuses on suffix based transformations which are common in English.

(3). Concept of Add delete Table -

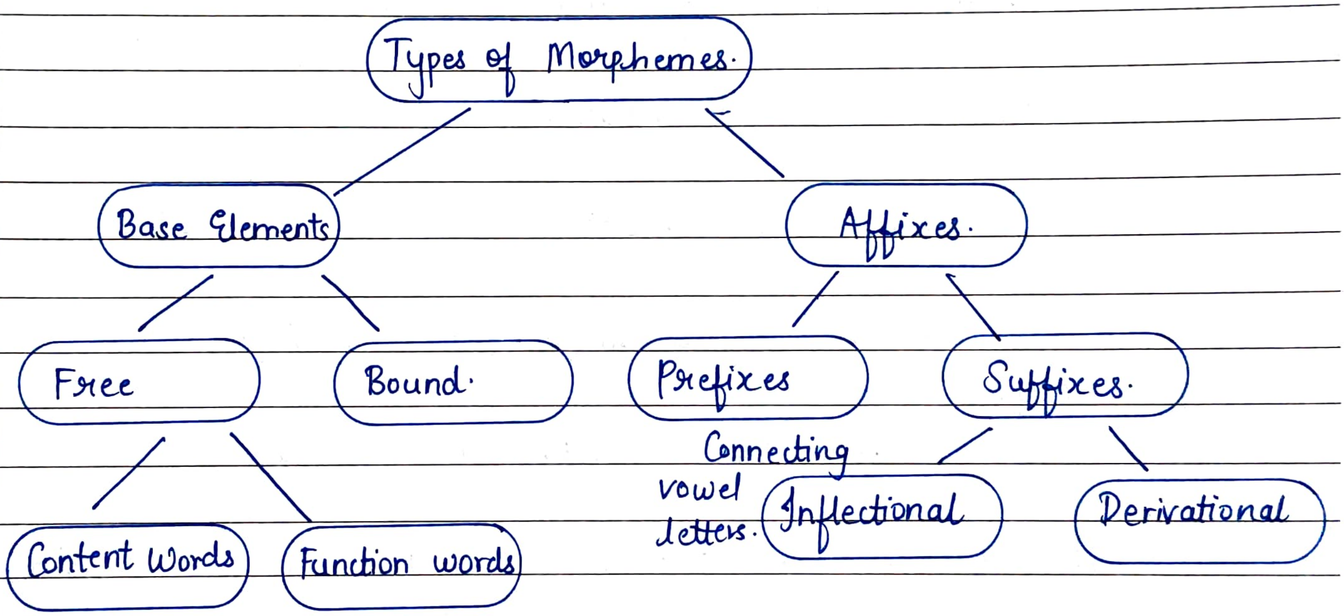
- The add delete table is a rule based representation that captures morphological variations using two simple operations.

  - Delete: → remove a portion of the original word.
  - Add → attach a new prefix or suffix.

- Thus a new word form is produced from the base word while maintaining semantic connection.

This approach is widely used in early NLP systems and classical stemming algorithms such as the Porter Stemmer, where suffix stripping follows pre defined add delete rules.

It is widely used in -
- Stemming algorithms.
- Rule based morphological analyzers.
- Information retrieval systems.



(4) <u>Structure of add delete Table -</u>

<u>The table usually contains -</u>

Original Word.	Delete	Add	Resultant word.
play	–	ing	playing
walk	–	ed	walked
read	–	ing	reading.

Here suffixes are added without deletion -

- Example 2 - Word Transformation with deletion -

Original	Delete	Add	Result
study	y	ies	Studies
carry	y	ied	carried
happy	y	ier	happier

The rule - remove y and add a new suffix.

(5). Mathematical representation -

Morphological transformation can be represented as -

$$\text{New Word} = (\text{Word} - \text{Delete}) + \text{Add}.$$

Example -

$$\text{studies} = (\text{study} - y) + \text{ies}.$$

The formal representation is useful in rule based NLP systems.

Let,

- W = Original word.
- D = Sequence to delete.
- A = Sequence to add.
- W' = Transformed word.

Rule is = $W' = (W - D) + A$.

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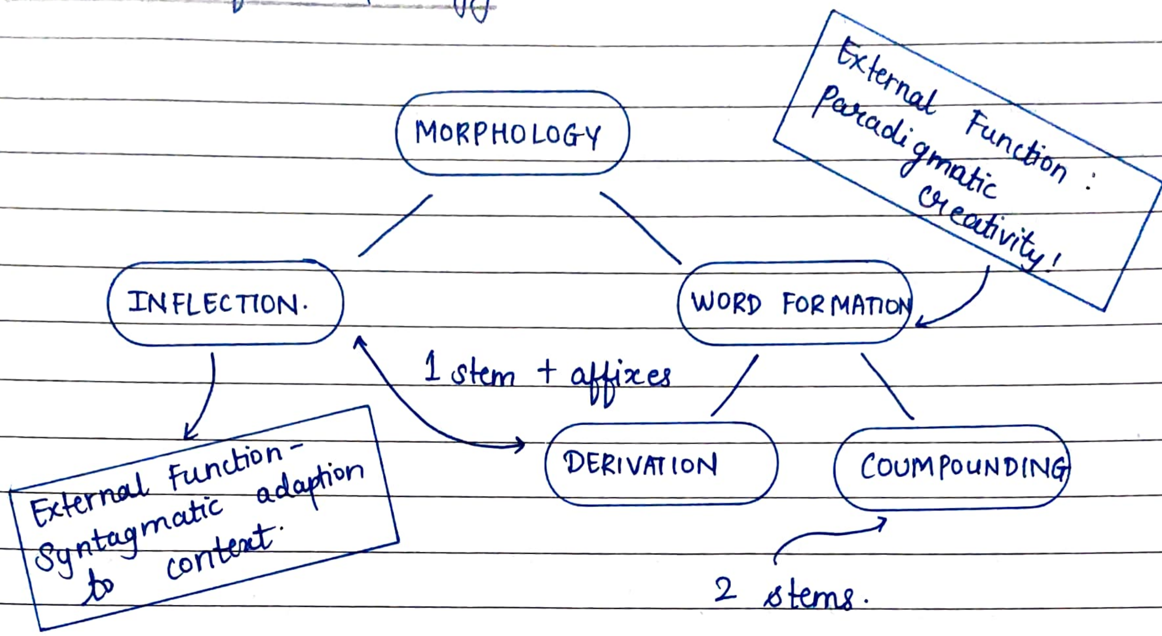
(6). Branches of Morphology -



(7) Applications in NLP -

- Stemming algorithms (porter stemmer)
- Lemmatization.
- Spell correction.
- Machine translation.
- Information retreival
- Text preprocessing.

(8) Advantages -

- Simple and interpretable.
- Easy rule creation.
- Useful for low resource languages.
- Improves text normalization.

(9) Limitations:

- Cannot handle irregular words easily (go - went)
- requires many rules.
- language dependent.
- Not robust for complex morphology.

* CONCLUSION:

- Morphology plays a crucial role in understanding language structure. The Add delete table provides a simple rule based mechanism to model how words change from through the addition and removal of morphemes. This approach, is foundational in NLP pre-processing and supports tasks such as stemming, lemmatization and grammar analysis by systematically representing word transformations.

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